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Kim et al.

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(54) **PUMP AND DISHWASHER COMPRISING THE SAME**

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(51) **Int. Cl.**

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A47L 15/14	(2006.01)
F04D 29/58	(2006.01)

(52) **U.S. Cl.**

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(Continued)

(58) **Field of Classification Search**

CPC A47L 15/4225; A47L 15/4234; A47L 15/00–508

See application file for complete search history.

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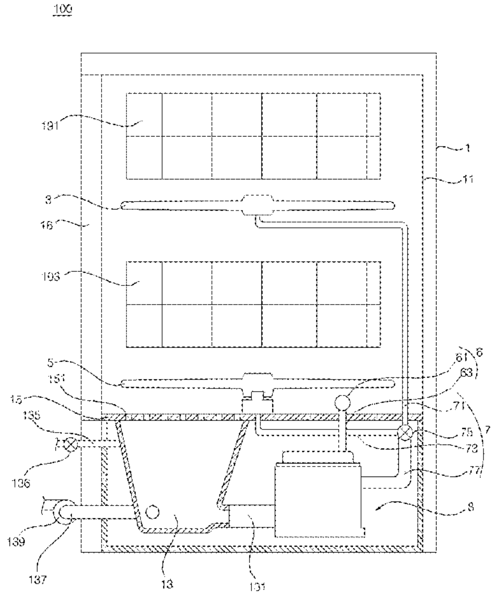
(57) **ABSTRACT**

A pump includes a housing, a water inlet pipe coupled to the housing and configured to receive washing water, a water outlet pipe coupled to the housing and configured to discharge washing water, an impeller located in the housing and configured to cause washing water to be transferred from the water inlet pipe to the water outlet pipe, a heater that is coupled to the housing, that is configured to heat washing water in the housing, and that is configured to generate steam based on heating washing water, and a steam discharge pipe coupled to the water outlet pipe and configured to discharge steam generated by the heater.

18 Claims, 10 Drawing Sheets



(52) **U.S. Cl.**
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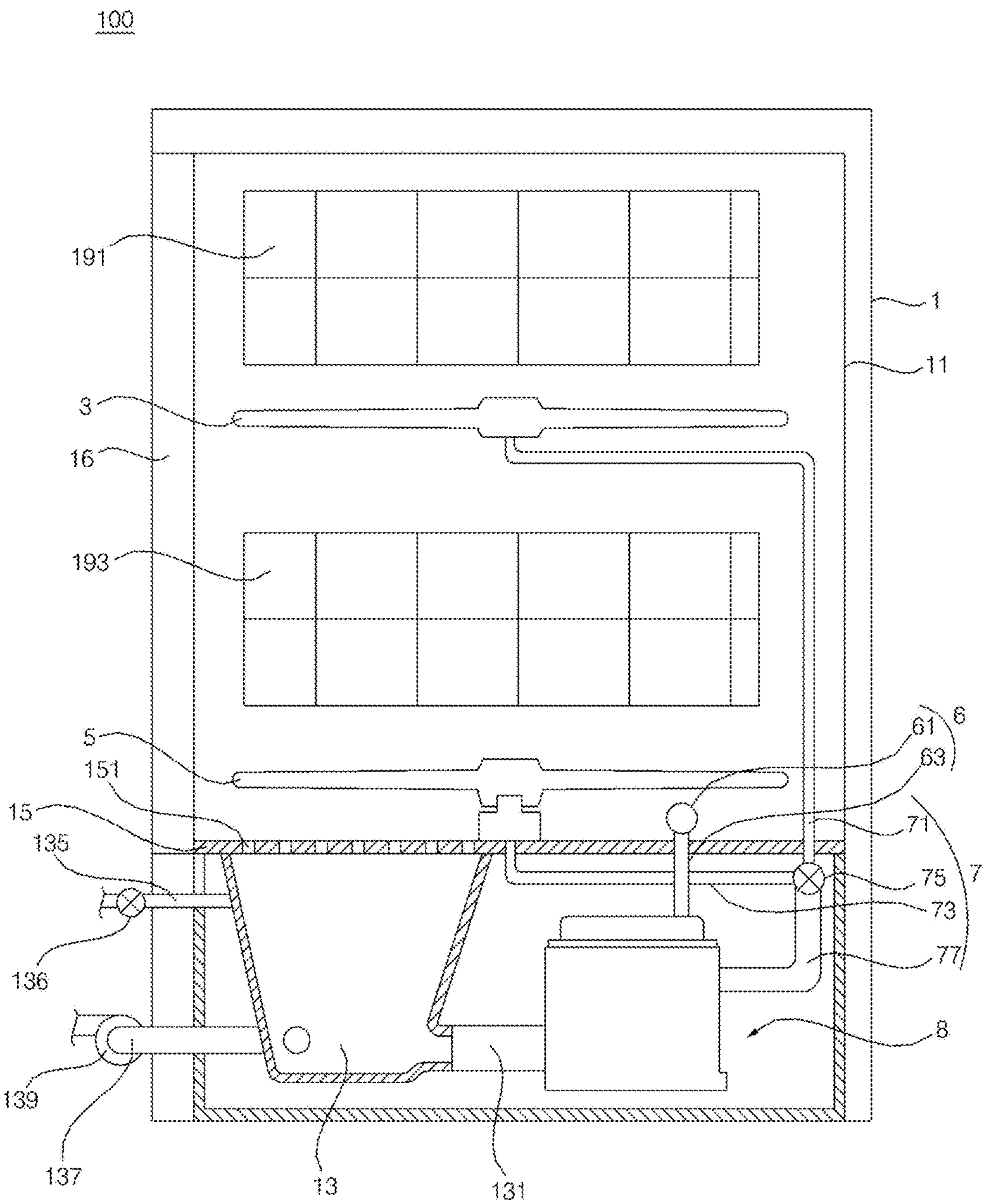
FIG. 1


100
191
1
11
3
16
193
6
61
63
5
151
15
71
135
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7
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8
139
137
13
131

FIG. 2

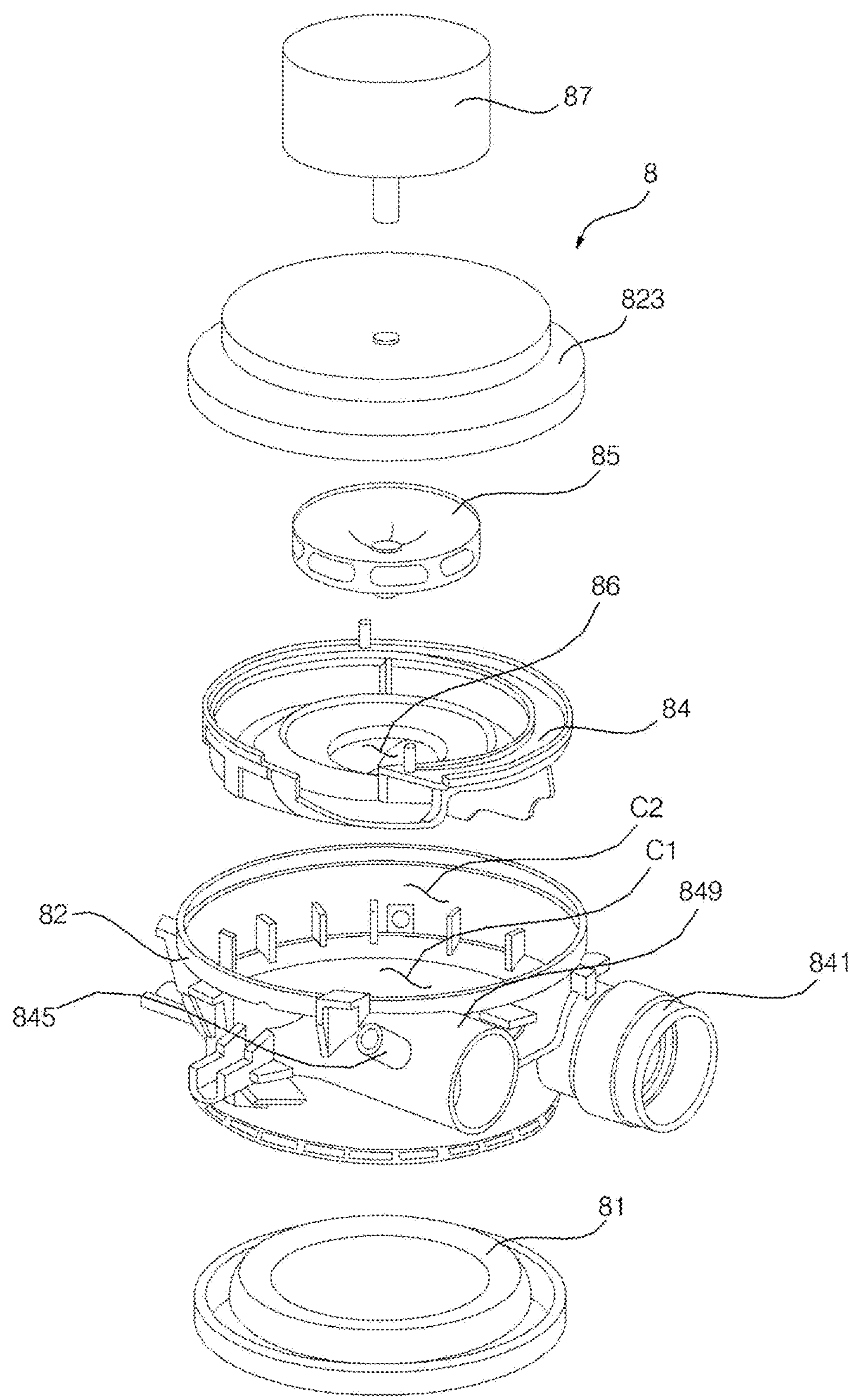


FIG. 3

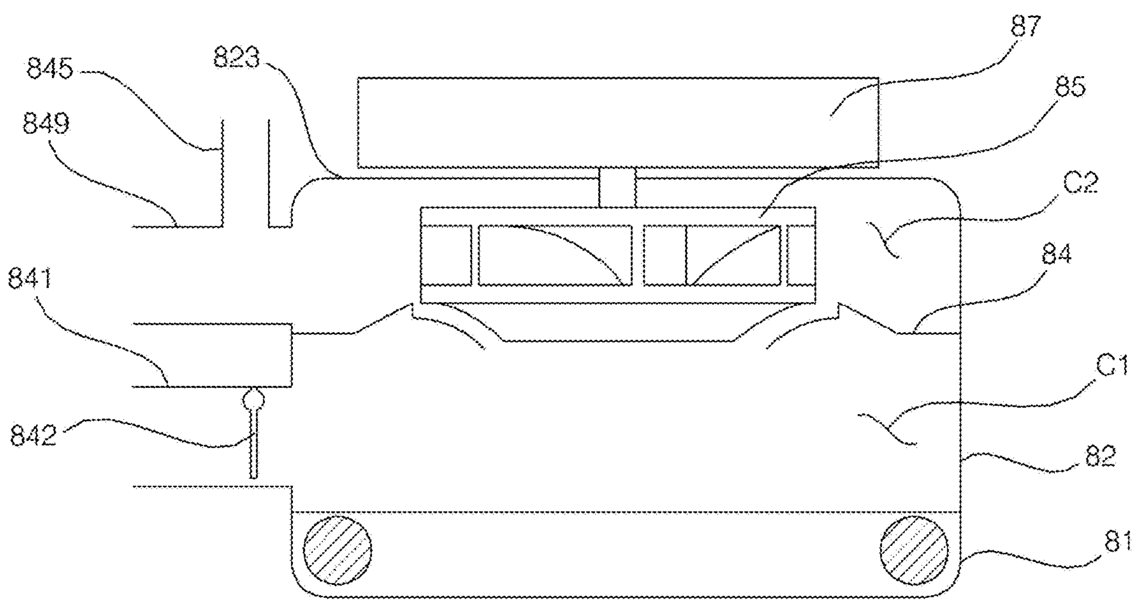


FIG. 4

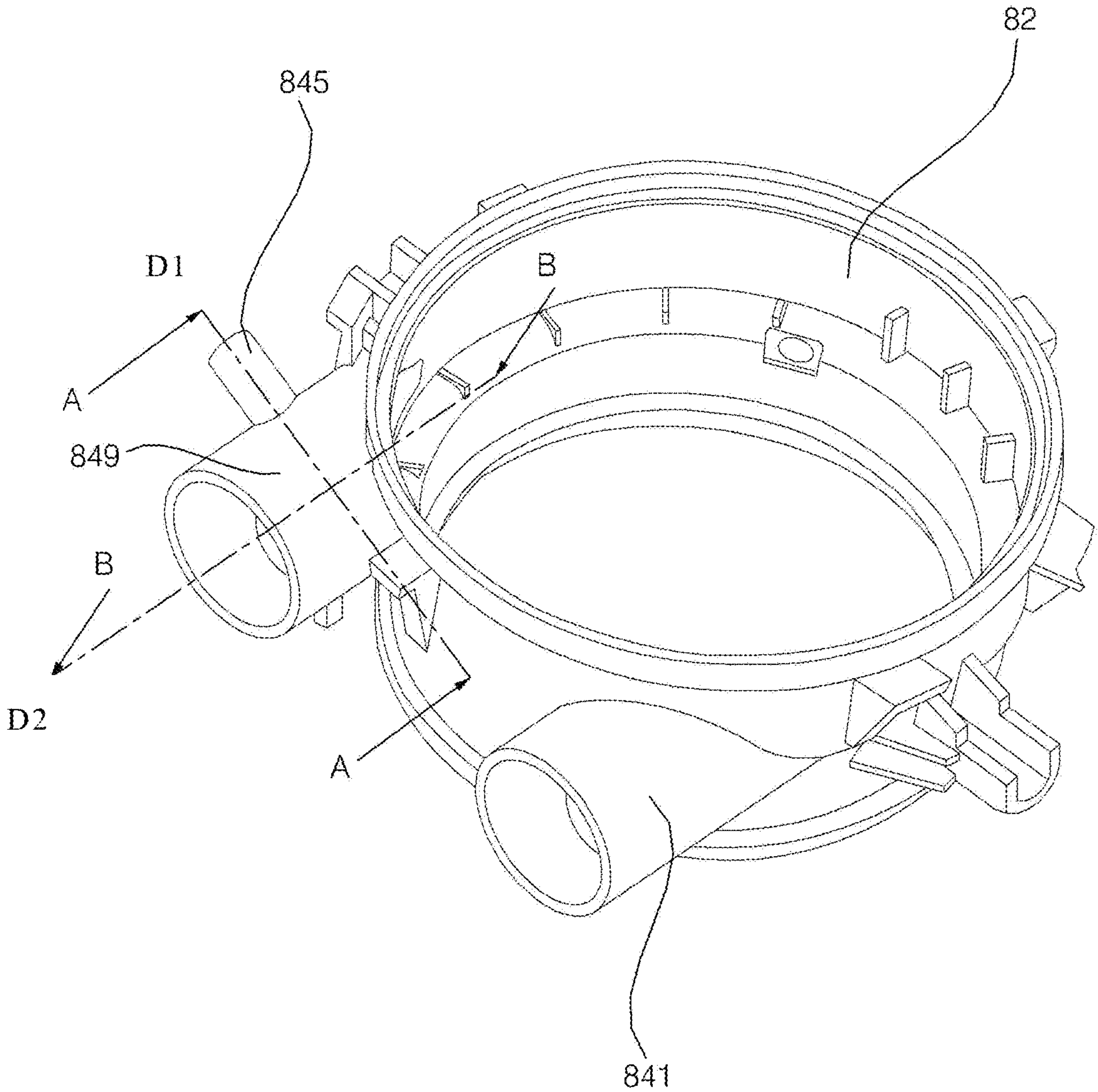


FIG. 5

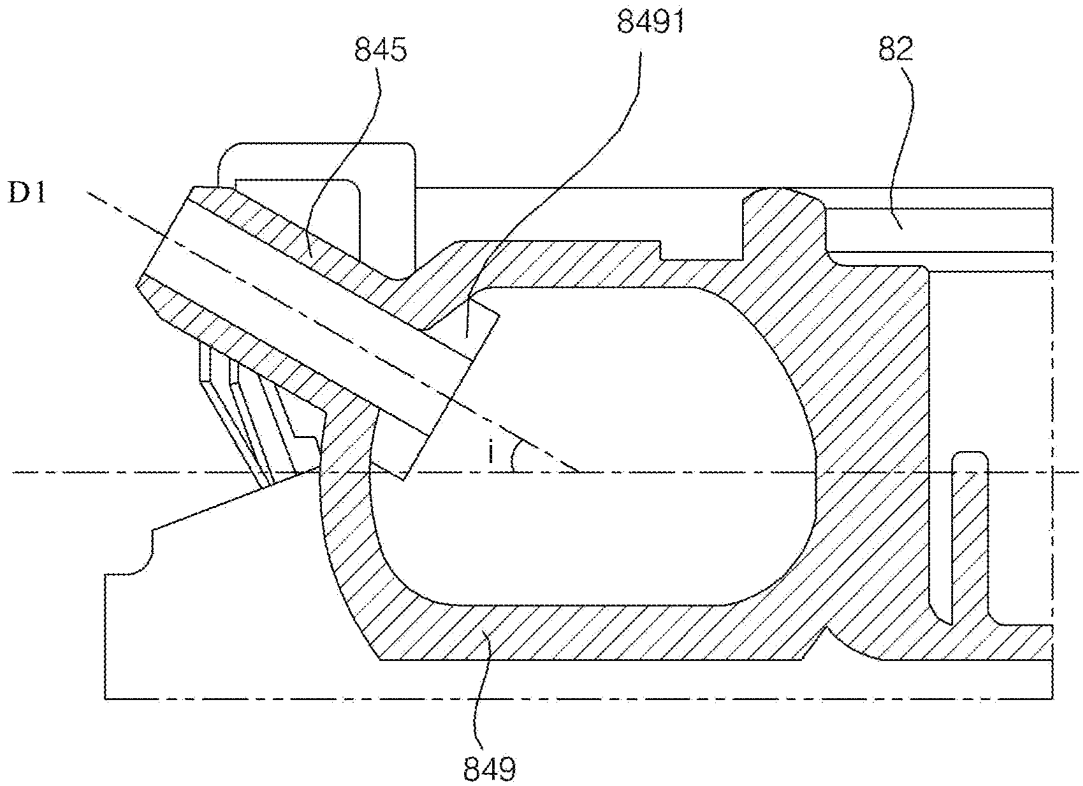


FIG. 6

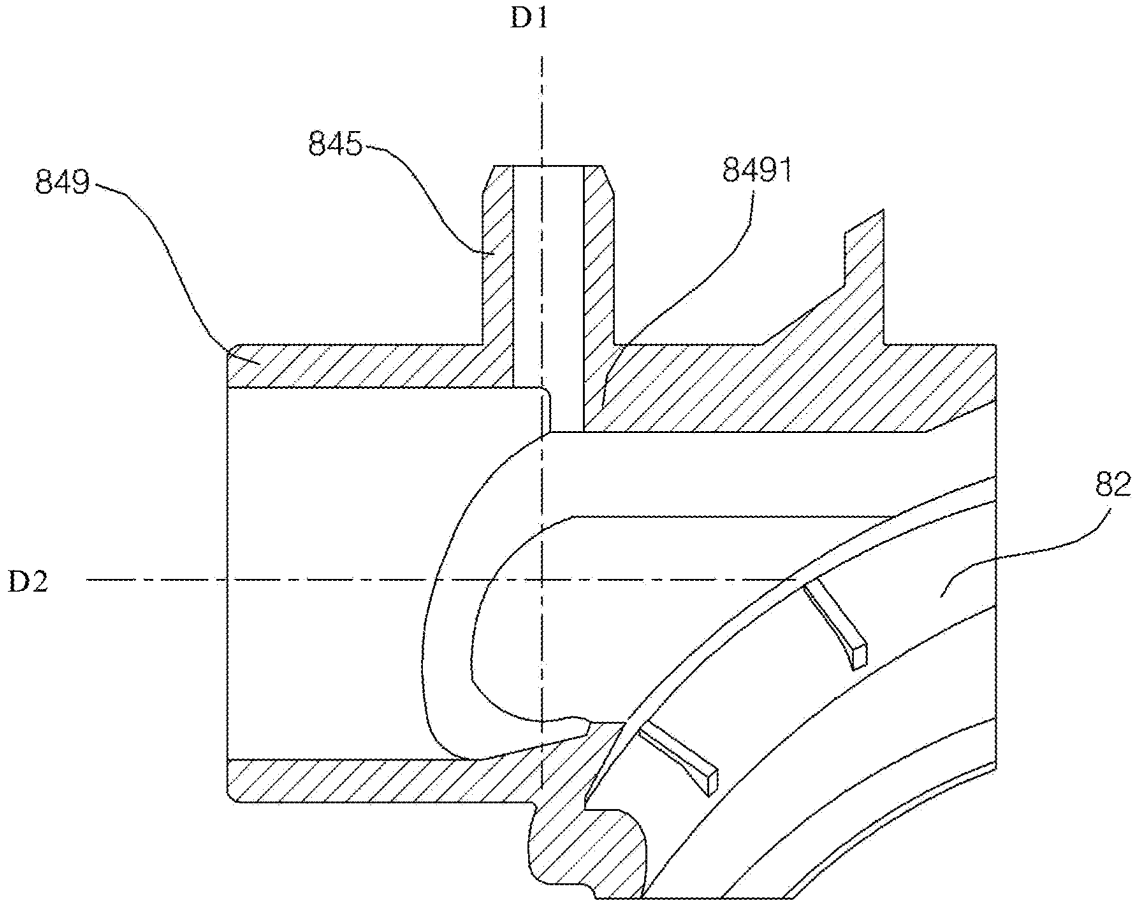


FIG. 7A

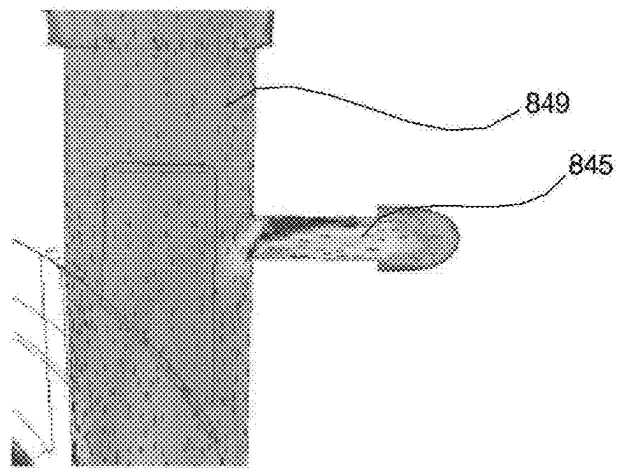


FIG. 7B

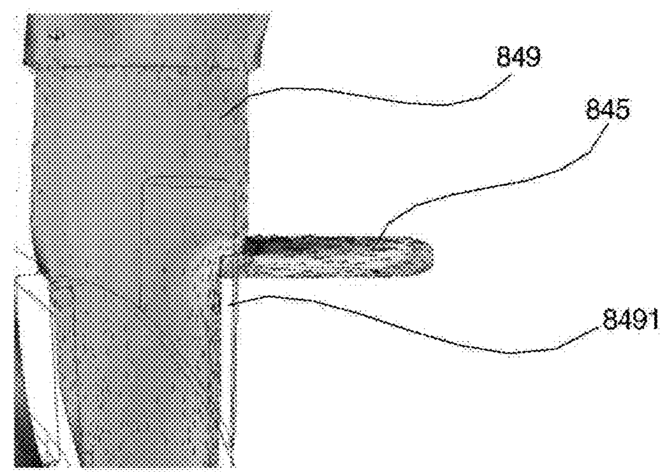


FIG. 8

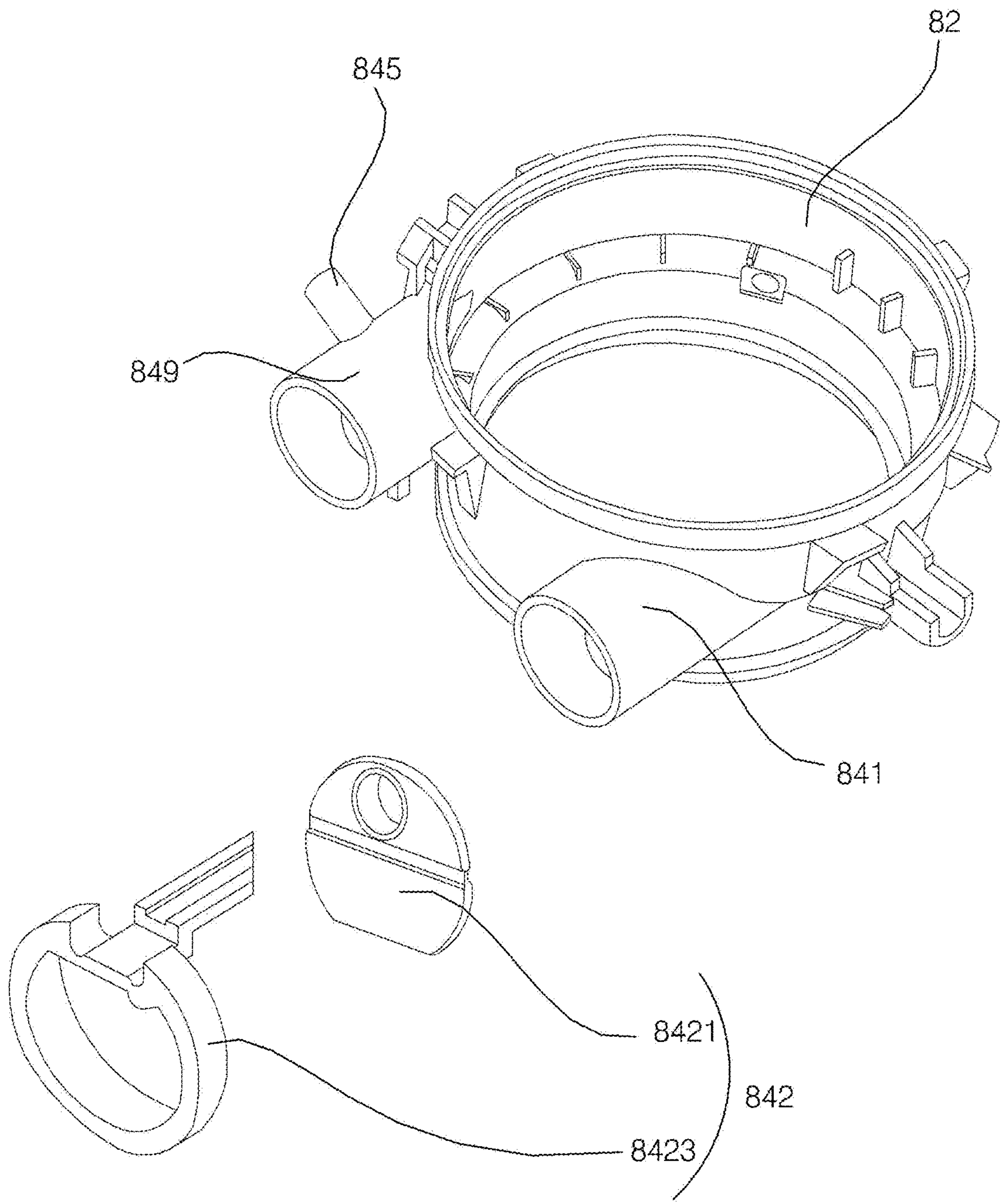


FIG. 9

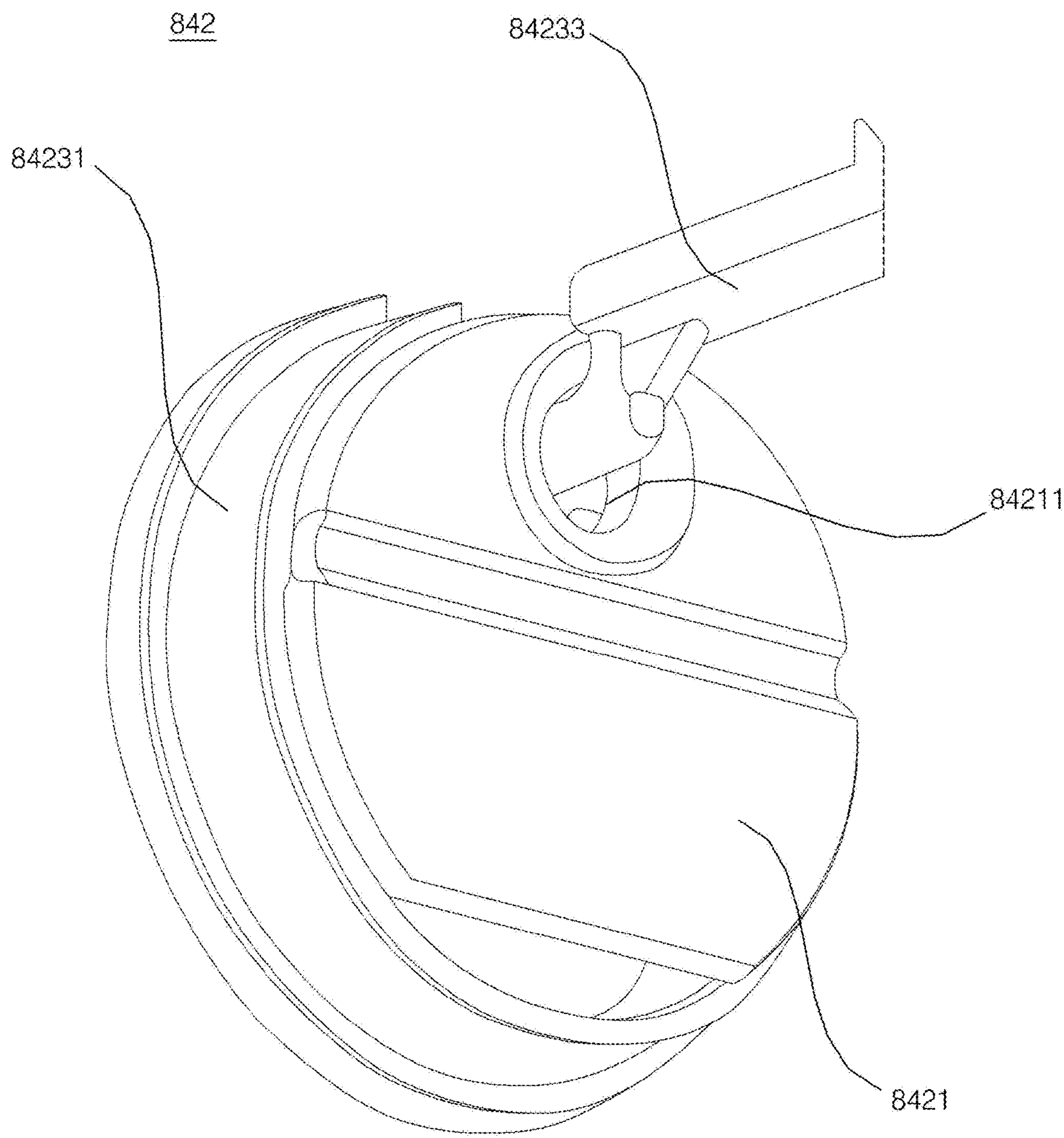
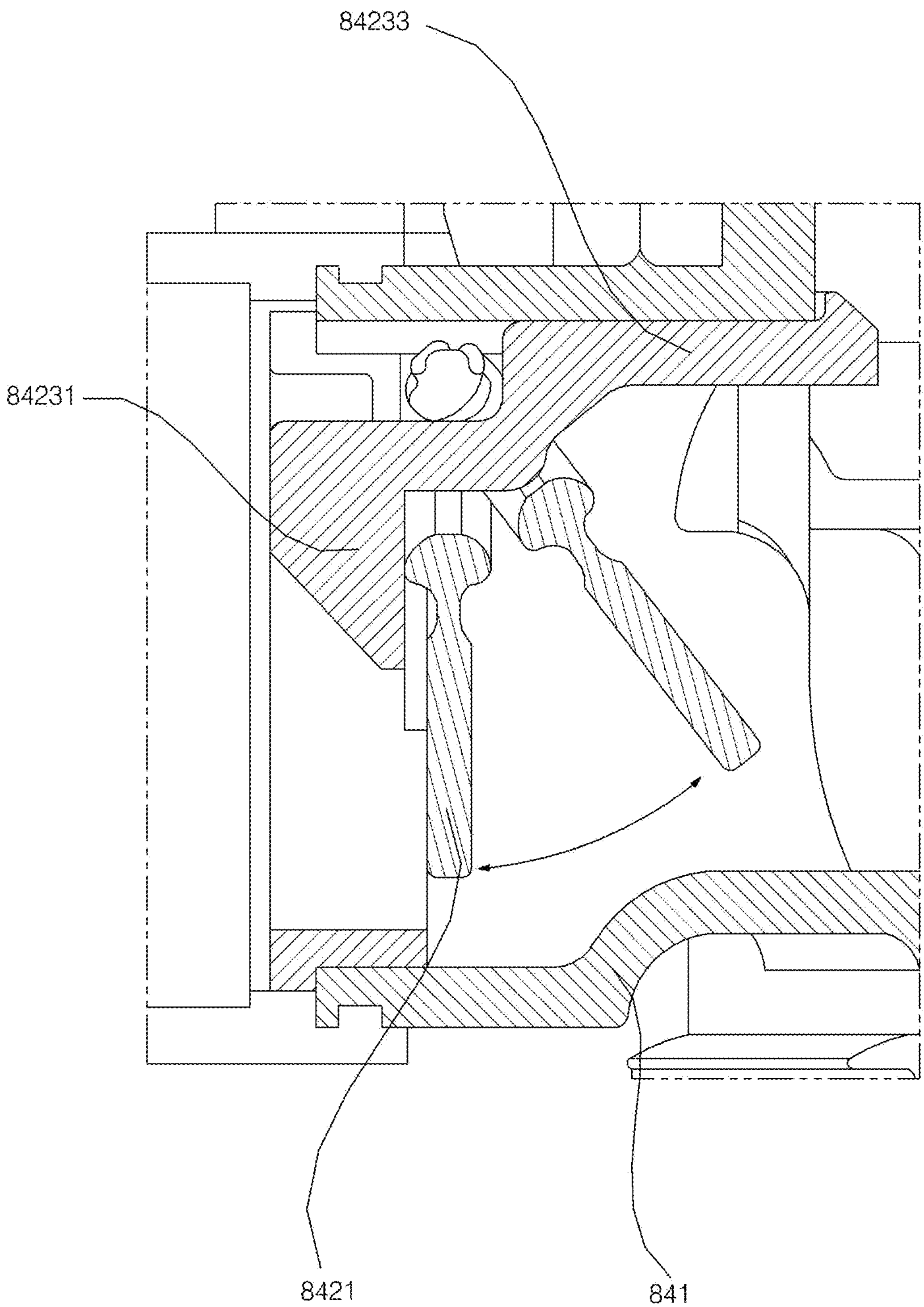


FIG. 10



PUMP AND DISHWASHER COMPRISING THE SAME

CROSS-REFERENCE TO RELATED APPLICATION

This application is a continuation of U.S. application Ser. No. 15/926,036, filed on Mar. 20, 2018, which claims the priority benefit of Korean Patent Application No. 10-2017-0034510, filed on Mar. 20, 2017, in the Korean Intellectual Property Office, the disclosures of which are incorporated herein by reference in their entirety.

FIELD

The present disclosure relates to a pump and a dishwasher including the same, and more particularly, to a pump that transmits washing water and that generates steam, and to a dishwasher including the pump.

BACKGROUND

A dishwasher is an appliance that can remove foreign matter on tableware, for example, by spraying washing water to the tableware. The dishwasher may include a tub that defines a washing space, a rack that is located in the tub and that accommodates tableware, a spraying arm that sprays washing water to the rack, a sump that stores washing water, and a pump that supplies washing water stored in the sump to the spraying arm.

In some examples, the dishwasher performs cleaning by using heated washing water, or performs cleaning or sterilization of tableware by supplying steam to the tableware. For example, dishwashers may generate hot water or steam by heating washing water stored in the sump through a heater provided inside the sump. In some cases where the heater is exposed to the inside of the sump and contacts washing water to heat washing water, the water level inside the sump may be controlled for the heater not to be exposed to prevent overheating of the heater. In some cases where heat transmission occurs when the heater contacts washing water, foreign matter may adhere to a surface of the heater in which a heat exchange efficiency may decrease and the surface of the heater may be corroded, which decreases the durability.

In some examples, a dishwasher may include a heater in a pump. In this case, the heater in the pump heats washing water to generate hot water or steam. In some cases, the performance of the pump may be deteriorated as air can enter a passage for discharging steam, or the steam performance may be deteriorated as steam can discharge through a passage for flow of washing water.

SUMMARY

The present disclosure provides a pump and a dishwasher that can generate steam and smoothly discharge steam without deteriorating the performance of the pump.

According to one aspect of the subject matter described in this application, a pump includes a housing, a water inlet pipe coupled to the housing and configured to receive washing water, a water outlet pipe coupled to the housing and configured to discharge washing water, an impeller located in the housing and configured to cause washing water to be transferred from the water inlet pipe to the water outlet pipe, a heater that is coupled to the housing, that is configured to heat washing water in the housing, and that is

configured to generate steam based on heating washing water, and a steam discharge pipe coupled to the water outlet pipe and configured to discharge steam generated by the heater.

Implementations according to this aspect may include one or more of the following features. For example, a cross-sectional area of the steam discharge pipe is less than a cross-sectional area of the water outlet pipe. The water outlet pipe may extend in a horizontal direction, and the steam discharge pipe may extend from the water outlet pipe in a direction that is inclined upward with respect to the horizontal direction.

In some implementations, the water outlet pipe may extend in a first direction, and the steam discharge pipe extends from the water outlet pipe in a second direction that is orthogonal to the first direction. The housing may define an internal space at an inside of the housing, and the pump further includes a partition wall that divides the internal space into a positive pressure chamber that is configured to accommodate the impeller and that is coupled to the water outlet pipe, and a negative pressure chamber that is coupled to water inlet pipe.

In some implementations, the water outlet pipe may include a stepped portion that protrudes from an inner surface of the water outlet pipe in which the stepped portion is located at a coupling region connected to the steam discharge pipe. In some cases, the water outlet pipe includes a first portion that extends from the coupling region toward the impeller, and a second portion that extends from the coupling region in a direction away from the impeller, where the stepped portion protrudes from an inner surface of the first portion of the water outlet pipe. The stepped portion may be located at an entrance of the steam discharge pipe. The pump may further include a check valve configured to close the water inlet pipe to restrict discharge of steam through the water inlet pipe.

According to another aspect, a dishwasher includes a tub, a spray arm configured to spray washing water into the tub, a steam supply unit configured to discharge steam into the tub, a sump configured to store washing water, and a pump that is configured to supply washing water stored in the sump to the spray arm, that is configured to generate steam, and that is configured to supply steam to the steam supply unit. The pump defines a negative pressure chamber that is coupled to the sump, and a positive pressure chamber that is coupled to the spray arm and the steam supply unit.

Implementations according to this aspect may include one or more of the following features. For example, the pump includes a partition wall that divides an internal space of the pump into the negative pressure chamber and the positive pressure chamber. In some examples, the pump includes a water outlet pipe coupled to the positive pressure chamber and configured to discharge washing water to the spray arm, and a steam discharge pipe coupled to the water outlet pipe and configured to discharge steam to the steam supply unit.

In some implementations, a cross-sectional area of the steam discharge pipe may be less than a cross-sectional area of the water outlet pipe. In some examples, the water outlet pipe extends in a horizontal direction, and the steam discharge pipe extends from the water outlet pipe in a direction that is inclined upward with respect to the horizontal direction. In some examples, the water outlet pipe extends in a first direction, and the steam discharge pipe extends from the water outlet pipe in a second direction that is orthogonal to the first direction.

In some implementations, the water outlet pipe includes a coupling region connected to the steam discharge pipe, a first portion that extends from the coupling region toward the positive pressure chamber, and a second portion that extends from the coupling region in a direction away from the positive pressure chamber, wherein a cross-sectional area of the first portion is less than a cross-sectional area of the second portion.

In some implementations, the pump includes a water inlet pipe coupled to the negative pressure chamber and configured to receive washing water from the sump, and a check valve configured to close the water inlet pipe to restrict discharge of steam through the water inlet pipe. The water outlet pipe may include a stepped portion that protrudes from an inner surface of the water outlet pipe, the stepped portion being located at a coupling region connected to the steam discharge pipe.

In some implementations, the check valve includes a valve body that is configured to rotate toward the negative pressure chamber based on supply of washing water through the water inlet pipe. In some examples, the valve body defines a cut portion that is spaced apart from an inner surface of the water inlet pipe and that allows washing water to discharge from the pump to the sump in a state in which the check valve closes the water inlet pipe.

BRIEF DESCRIPTION OF THE DRAWINGS

The objects, features and advantages of the present disclosure will be more apparent from the following detailed description in conjunction with the accompanying drawings.

FIG. **1** is a schematic cross-sectional view showing an example dishwasher.

FIG. **2** is an exploded perspective view showing an example pump.

FIG. **3** is a schematic cross-sectional view showing an example pump.

FIG. **4** is a perspective view showing a portion of an example pump. FIG. **5** is a cross-sectional view taken along line A-A of FIG. **4**.

FIG. **6** is a cross-sectional view taken along line B-B in FIG. **4**.

FIGS. **7**A and **7**B illustrate example experimental results of washing water outflow of an example pump.

FIG. **8** is a partial exploded perspective view showing an example pump.

FIG. **9** is a perspective view showing an example check valve of an example pump.

FIG. **10** is an operational example of an example check valve of an example pump.

DETAILED DESCRIPTION

Exemplary implementations of the present disclosure are described with reference to the accompanying drawings in detail. The same reference numbers are used throughout the drawings to refer to the same or like parts. Detailed descriptions of well-known functions and structures incorporated herein may be omitted to avoid obscuring the subject matter of the present disclosure.

Hereinafter, the present disclosure will be described with reference to the drawings for explaining a pump and a dishwasher including the same according to implementations of the present disclosure.

FIG. **1** illustrates an example dishwasher.

A dishwasher **100** may include a cabinet **1** that defines an outer appearance, a tub **11** located inside the cabinet **1** and configured to accommodate a dishware, a plurality of spraying arms **3** and **5** configured to spray washing water into the tub **11**, a steam supply unit **6** configured to discharge steam into the tub **11**, a sump **13** configured to store washing water, and a pump **8** that is configured to supply the washing water stored in the sump **13** to the plurality of spraying arms **3** and **5** and that is configured to generate steam and supply steam to the steam supply unit **6**.

A plurality of racks **191** and **193** for storing dishware may be located in the tub **11**. The plurality of racks **191** and **193** may include an upper rack **191** provided in an upper area of the tub **11** and a lower rack **193** provided in a lower area of the tub **11**.

The tub **11** may be opened and closed by a door **16** provided on one side of the cabinet. A user may take out the plurality of racks **191** and **193** from the tub **11** after opening the door **16**.

For example, the plurality of spraying arms may include an upper spraying arm **3** for spraying washing water to the upper rack **191** and a lower spraying arm **5** for spraying washing water to the lower rack **193**. The washing water sprayed from the plurality of spraying arms **3** and **5** may be collected into the sump **13**.

The sump **13** may be provided in a lower portion of the tub **11** to store washing water. A sump cover **15** may be disposed in an upper side of the sump **13**. The sump cover **15** may be provided with a recovery hole **151** so that washing water in the tub **11** can pass through the recovery hole **151** and flow into the sump **13**.

The sump **13** may be coupled to an external water source through a water supply channel **135**. The water supply channel **135** may be opened and closed by a water supply valve **136**. The washing water stored in the sump **13** may be discharged to the outside of the cabinet **1** through a drainage pipe **137** and a drain pump **139**.

The washing water stored in the sump **13** may be supplied to the plurality of spraying arms **3** and **5** through the pump **8** and a washing water supply unit **7**. The washing water supply unit **7** may include a main connection pipe **77** coupled to the pump **8**, an upper portion connection pipe **71** connecting the main connection pipe **77** and the upper spraying arm **3**, and a lower portion connection pipe **73** connecting the main connection pipe **77** and the lower spraying arm **5**.

The upper portion connection pipe **71** and the lower portion connection pipe **73** may be branched from the main connection pipe **77**, and a switching valve **75** for controlling the opening and closing of the upper portion connection pipe and/or the lower portion connection pipe **73** may be provided in a branch point of the upper portion connection pipe **71** and the lower portion connection pipe **73**.

The sump **13** may be coupled to the pump **8** and a supply pipe **131**. The supply pipe **131** may guide the washing water in the sump **13** to the pump **8**.

The steam supply unit **6** may discharge steam into the tub **11**. The steam discharged from the steam supply unit **6** may rise and act on the tableware housed in the plurality of racks **191** and **193**. The steam supply unit **6** may include a steam nozzle **61** disposed in the tub **11** to discharge steam and a steam supply pipe **63** connecting the steam nozzle **61** and the pump **8**.

The pump **8** may transmit the washing water stored in the sump **13** to the washing water supply unit **7**. The pump **8** may heat the washing water transmitted to the washing water supply unit **7**. The pump **8** may suck the washing water stored in the sump **13** and supply to the plurality of spraying arms **3** and **5** through the washing water supply unit **7**.

The pump **8** may generate steam and supply it to the steam supply unit **6**. The pump **8** may generate steam by heating the washing water stored in the pump **8** without transmitting the washing water. The steam generated in the pump **8** may be discharged into the tub **11** through the steam supply unit **6**.

The pump **8** will be described in detail with reference to FIG. **2** and following drawings.

FIG. **2** is an exploded perspective view showing an example pump, and FIG. **3** is a schematic cross-sectional view of an example pump.

For example, the pump **8** may include a housing **82** having a cylindrical shape which is fixed inside the cabinet **1**, a washing water inlet pipe **841** which is coupled to the housing and into which the washing water flows, a washing water outlet pipe **849** which is coupled to the housing **82** and discharges the washing water, an impeller **85** which is disposed inside the housing **82** to transfer the washing water from the washing water inlet pipe **841** to the washing water outlet pipe **849**, a heater **81** coupled to the housing **82** and heats the washing water inside the housing **82** to generate steam, and a steam discharge pipe **845** which is coupled to the washing water outlet pipe **849** and discharges the steam generated by the heater **81**.

The housing **82** may be formed in a cylindrical shape having open top and bottom. A housing cover **823** may be coupled to the upper end of the housing **82**, and the heater may be coupled to the lower end thereof. The housing cover **823** may cover the upper portion of the housing **82**, and the housing cover **823** may be provided with a motor **87** for generating a rotational force to rotate the impeller **85**.

A partition wall **84** may be disposed inside the housing **82**. The partition wall **84** may divide an internal space of the housing **82** into upper and lower portions. The partition wall **84** may form a negative pressure chamber C**1** and a positive pressure chamber C**2** inside the housing **82**. The negative pressure chamber C**1** may be a place where a negative pressure is generated by the rotation of the impeller **85** and the positive pressure chamber C**2** may be a place where a positive pressure is generated by the rotation of the impeller **85**. The negative pressure chamber C**1** may be coupled to the sump **13** through the supply pipe **131** and the washing water inlet pipe **841**. The positive pressure chamber C**2** may be coupled to the plurality of spraying arms **3**, **5** through the washing water outlet pipe **849** and the washing water supply unit **7**.

In some implementations, the partition wall **84** may define a communication hole **86** configured to communicate with the negative pressure chamber C**1** and the positive pressure chamber C**2**. The surface of the partition wall **84** in the positive pressure chamber C**2** side may be formed in a volute.

The impeller **85** may be rotated by the motor **87** to discharge the water in the housing **82** to the outside. The impeller **85** may be rotatably disposed in the positive pressure chamber C**2**. The impeller **85** may transfer the washing water introduced from the negative pressure chamber C**1** into the positive pressure chamber C**2** through the communication hole **86** to the washing water outlet pipe **849**.

The heater **81** may be coupled to the lower end of the housing **82** to form a bottom surface of the housing **82**. The heater **81** may heat the washing water flowing inside the housing **82** when the impeller **85** rotates. The heater **81** may generate steam by heating the washing water in the housing **82** when the impeller **85** stops.

The washing water inlet pipe **841** may be coupled to the negative pressure chamber C**1** side of the housing **82**. The washing water inlet pipe **841** may be coupled to the supply pipe **131** and the washing water of the sump **13** may flow into the negative pressure chamber C**1**. The washing water inlet pipe **841** may be disposed to protrude outward from the lower side wall of the housing **82**. The washing water inlet pipe **841** may be arranged in such a manner that a pipe direction is horizontal, so that the washing water flows in the horizontal direction. A check valve **842** for opening and closing the washing water inlet pipe **841** is disposed in the washing water inlet pipe **841**. Details of the check valve **842** will be described later with reference to FIG. **8** to FIG. **10**.

The washing water outlet pipe **849** may be coupled to the positive pressure chamber C**2** side of the housing **82**. The washing water outlet pipe **849** may be coupled to the main connection pipe **77** of the washing water supply unit **7** so that the washing water of the positive pressure chamber C**2** flows out to the main connection pipe **77**. The washing water outlet pipe **849** may be arranged to protrude outward from the upper side wall of the housing **82**. The washing water outlet pipe **849** may be arranged in such a manner that a pipe direction is horizontal, so that the washing water flows in the horizontal direction.

The steam discharge pipe **845** is coupled to the positive pressure chamber C**2** side of the housing **82**. The steam discharge pipe **845** may be disposed in various locations of the housing **82** so as to connect the positive pressure chamber C**2** of the housing **82** and the steam supply unit **6**. In the present implementation, the steam discharge pipe **845** is coupled to the washing water outlet pipe **849**. The steam discharge pipe **845** may discharge steam generated in the housing **82** by the heater **81** to the steam supply unit **6**. The steam supply unit **6** may be coupled to the positive pressure chamber C**2** through the steam discharge pipe **845**.

FIG. **4** is a perspective view of a portion of an example pump, FIG. **5** is a cross-sectional view taken along line A-A of FIG. **4**, FIG. **6** is a cross-sectional view taken along line B-B in FIG. **4**, and FIGS. **7**A and **7**B illustrate example experimental results of washing water outflow of an example pump.

The cross sectional area of the steam discharge pipe **845** may be smaller than the cross sectional area of the washing water outlet pipe **849**. The cross sectional area of the steam discharge pipe **845** may be formed to be smaller than the cross sectional area of the washing water outlet pipe **849** to minimize the loss of washing water into the steam discharge pipe **845**.

The steam discharge pipe **845** may be arranged in such a manner that the pipe direction D**1** is inclined to the upward direction from the horizontal direction. As shown in FIG. **5**, the pipe direction D**1** of the steam discharge pipe **845** may have an inclination angle i between the horizontal direction and an acute angle. Since the steam in the housing **82** flows in a direction opposite to the direction of gravity, the pipe direction D**1** of the steam discharge pipe **845** may be inclined upward so that the loss of steam from the horizontal washing water outlet pipe **849** having a horizontal pipe direction D**2** to the washing water supply unit **7** can be minimized.

The steam discharge pipe **845** may be arranged to be orthogonal to the washing water outlet pipe **849**. That is, the steam discharge pipe **845** and the washing water outlet pipe **849** may be arranged in a "T" shape. As shown in FIG. **6**, the pipe direction D**1** of the steam discharge pipe **845** may be disposed to be perpendicular to the pipe direction D**2** of the washing water outlet pipe **849**, so that the loss of the washing water to the steam discharge pipe **845** can be minimized and the loss of steam to the washing water supply unit **7** can be minimized.

Referring to FIG. **6**, an upstream side cross-sectional area of the washing water outlet pipe **849** may be smaller than a downstream side cross-sectional area based on a portion where the steam discharge pipe **845** is connected. The above-described division between the upstream side and the downstream side may be based on the flow direction of the washing water.

The washing water outlet pipe **849** may include a stepped portion **8491** in which an upstream side, based on the flow direction of the washing water, in a portion where the steam discharge pipe **845** is connected protrudes inward. The stepped portion **8491** may be located at an entrance of the steam discharge pipe **845** based on the steam flow direction. The stepped portion **8491** may allow the washing water flowing in the washing water outlet pipe **849** to pass by the steam discharge pipe **845** without being expelled due to inertia.

FIG. **7**A shows a velocity distribution of the washing water when the stepped portion **8491** does not exist, and FIG. **7**B shows a velocity distribution of the washing water when the stepped portion **8491** exists. In FIGS. **7**A and **7**B, the dark regions in the steam discharge pipe **845** indicate a faster washing water speed, and the light regions in the steam discharge pipe **845** indicate a slower washing water speed. Based on the results of the experiment, when the stepped portion **8491** exists, the positive pressure of the steam discharge pipe **845** may be reduced, and the loss of the washing water to the steam discharge pipe **845** may be minimized.

FIG. **8** is a partial exploded perspective view of an example pump, FIG. **9** is a perspective view of an example check valve of an example pump, and FIG. **10** is an operational example of an example check valve of an example pump.

The check valve **842** may open and close the washing water inlet pipe **841** so that the steam which is generated inside the housing **82** by the heater **81** may not flow out through the washing water inlet pipe **841**. The check valve **842** may be opened when the impeller **85** rotates and may be closed when the impeller **85** stops. That is, the check valve **842** may be opened when the washing water flows and may be closed when the washing water does not flow. The check valve **842** may close the washing water inlet pipe **841** during the steam generation of the heater **81** so that the steam generated inside the housing **82** may not flow out to the supply pipe **131** through the washing water inlet pipe **841** from the negative pressure chamber C**1**.

The check valve **842** may include a valve body **8421** having a circular segment shape, and a valve fixture **8423**, which is coupled so that the valve body **8421** can be rotatable.

The valve body **8421** may be rotated to open and close the washing water inlet pipe **841**. The valve body **8421** may rotate toward the negative pressure chamber C**1** side when opened. The valve body **8421** may be provided with a valve hole **84211** through which a part of the valve fixture **8423** is passed.

The valve body **8421** may be formed in a circular segment shape so as not to completely block the washing water inlet pipe **841**. The valve body **8421** may be formed in a circular shape having a cut lower portion so that a part of the lower portion of the washing water inlet pipe **841** is opened when closed. In some cases where the washing water in the pump **8** should be drained, the drain pump **139** may operate to drain water from the pump **8**. For example, when the drain pump **139** operates, the washing water in the pump **8** may flow into the supply pipe **131** through a gap between the valve body **8421** and the washing water inlet pipe **841** and may be discharged to the outside through the sump **13** and the drainage pipe **137**.

The valve fixture **8423** may be coupled so that the valve body **8421** can rotate. The valve fixture **8423** may be coupled to the washing water inlet pipe **841** so that the valve body **8421** can be arranged to be rotatable for the washing water inlet pipe **841**. The valve fixture **8423** may be provided with a ring-shaped outlet end connecting unit **84231** and a hinge unit **84233** penetrating the valve hole **84211**.

The outlet end connecting unit **84231** may be formed in a ring shape and may be coupled to a distal end of the washing water inlet pipe **841**. The circumference of the outlet end connecting unit **84231** may be in contact with the valve body **8421**. The hinge unit **84233** may pass through the valve hole **84211** and may be coupled to the washing water inlet pipe **841**.

When the washing water is introduced through the supply pipe **131** during rotation of the impeller **85**, the valve body **8421** may be rotated by the flow pressure of the washing water to open the washing water inlet pipe **841**. When the impeller **85** is stopped, the valve body **8421** may be in contact with the outlet end connecting unit **84231** to close the washing water inlet pipe **841**.

Example operations of the pump and the dishwasher will be described as follows.

When the water supply valve **136** is opened, washing water may be supplied from an external water supply source through the water supply channel **135** and be stored in the sump **13**. When the motor **87** rotates the impeller **85**, the washing water stored in the sump **13** may flow into the washing water inlet pipe **841** through the supply pipe **131**. At this time, the check valve **842** may be rotated by the flow pressure of the washing water to open the washing water inlet pipe **841**. The washing water flowed into the washing water inlet pipe **841** may flow into the negative pressure chamber C**1** inside the housing **82**. The washing water in the negative pressure chamber C**1** may flow into the positive pressure chamber C**2** through the communication hole **86** and then flow into the washing water outlet pipe **849** by the rotation of the impeller **85**. At this time, the heater **81** may heat the washing water flowed into the housing **82**. The washing water flowed into the washing water outlet pipe **849** may partially flow to the steam discharge pipe **845** but mostly may flow to the washing water supply unit **7**. The washing water flowed into the washing water supply unit **7** may be sprayed through the upper spraying arm **3** and/or the lower spraying arm **5** depending on the operation of the switching valve **75**. The washing water sprayed through the upper spraying arm **3** and/or the lower spraying arm **5** may be recovered to the sump **13**.

In some implementations, when the heater **81** operates but the motor **87** does not rotate the impeller **85**, the washing water in the housing **82** may be heated to generate steam. The steam generated in the housing **82** may flow into the washing water outlet pipe **849** and then flow to the steam discharge pipe **845**. The steam flowing in the washing water outlet pipe **849** can be partially flow to the washing water supply unit **7** but mostly flow to the steam discharge pipe **845**.

The steam flowed into the steam discharge pipe **845** may be discharged into the tub **11** through the steam supply unit **6**.

According to the pump of the present disclosure and the dishwasher including the same, one or more of the following effects can be obtained.

For example, the steam discharge pipe coupled to the positive pressure chamber of the pump may prevent air in the tub from flowing into the pump, thereby securing the flow rate of the pump.

In some implementations, the steam discharge pipe coupled to the washing water outlet pipe may minimize flow of steam generated in the pump to the spraying arm side, thereby improving the steam performance.

In some implementations where the steam discharge pipe is coupled to the washing water outlet pipe, the pump performance may be improved by minimizing discharge of washing water to the steam discharge pipe.

In some implementations, steam generated in the pump is restricted from flowing out to the sump side by the check valve disposed in the washing water inlet pipe, thereby improving the steam performance.

Hereinabove, although the present disclosure has been described with reference to exemplary implementations and the accompanying drawings, the present disclosure is not limited thereto, but may be variously modified and altered by those skilled in the art to which the present disclosure pertains without departing from the spirit and scope of the present disclosure claimed in the following claims.

What is claimed is:

1. A dishwasher comprising:

a tub;

a spray arm configured to spray washing water into the tub;

a steam supplier configured to discharge steam into the tub;

a sump configured to store washing water; and

a pump that is configured to supply washing water stored in the sump to the spray arm, that is configured to generate steam, and that is configured to supply steam to the steam supplier, the pump comprising:

a housing configured to receive washing water from the sump,

a heater disposed below the housing and configured to heat washing water in the housing, the heater being configured to generate steam based on heating washing water in the housing,

a partition wall that divides an inside space of the housing into an upper portion and a lower portion, the partition wall defining a communication hole that enables fluid communication between the upper portion and the lower portion of the housing,

an impeller disposed in the upper portion of the housing, and

a water outlet pipe that protrudes from the upper portion of the housing, that communicates with the housing, and that is configured to discharge washing water from the housing,

wherein the water outlet pipe defines a steam discharge hole at a position above a horizontal axis passing through a circumferential surface of the water outlet pipe, and

wherein a center axis of the steam discharge hole defines an acute angle with respect to the horizontal axis.

2. The dishwasher of claim **1**, wherein the pump further comprises a steam discharge pipe that is located at the steam discharge hole, that protrudes from the circumferential surface of the water outlet pipe, and that is configured to carry steam discharged through the steam discharge hole.

3. The dishwasher of claim **1**, wherein the steam discharge hole is spaced apart from the housing.

4. The dishwasher of claim **1**, wherein the water outlet pipe protrudes from a circumferential surface of the housing, and

wherein the pump further comprises a water inlet pipe that protrudes from the circumferential surface of the housing, that is coupled to the lower portion of the housing, and that is configured to provide washing water to the housing.

5. The dishwasher of claim **4**, wherein the water outlet pipe and the water inlet pipe extend parallel to each other, and are disposed at a first side of the housing with respect to a rotational axis of the impeller.

6. The dishwasher of claim **1**, wherein the water outlet pipe includes a stepped portion that protrudes from an inner surface of the water outlet pipe and that faces the steam discharge hole.

7. The dishwasher of claim **6**, wherein the stepped portion is located at one side of the steam discharge hole.

8. The dishwasher of claim **6**, wherein the stepped portion is located at an upstream side relative to the steam discharge hole, the upstream side facing the housing.

9. The dishwasher of claim **8**, wherein a diameter of the water outlet pipe at the upstream side is less than a diameter of the water outlet pipe at a downstream side relative to the steam discharge hole.

10. The dishwasher of claim **1**, wherein a cross-sectional area of the steam discharge hole is less than a cross-sectional area of the water outlet pipe.

11. The dishwasher of claim **1**, wherein the center axis of the steam discharge hole is orthogonal to a longitudinal axis extending along the water outlet pipe.

12. The dishwasher of claim **1**, wherein the pump further comprises a steam discharge pipe that protrudes from the circumferential surface of the water outlet pipe and that extends along the center axis of the steam discharge hole, the steam discharge pipe being configured to carry steam discharged through the steam discharge hole.

13. The dishwasher of claim **1**, wherein the pump further comprises a steam discharge pipe that is connected to the circumferential surface of the water outlet pipe and that extends upward from the steam discharge hole.

14. A dishwasher comprising:

a tub;

a spray arm configured to spray washing water into the tub;

a steam supplier configured to discharge steam into the tub;

a sump configured to store washing water; and

a pump that is configured to supply washing water stored in the sump to the spray arm, that is configured to generate steam, and that is configured to supply steam to the steam supplier, the pump comprising:

a housing configured to receive washing water from the sump,

a heater configured to heat washing water in the housing and to generate steam based on heating washing water in the housing,

an impeller disposed in the housing, and

a water outlet pipe that protrudes from the housing, that communicates with the housing, and that is configured to discharge washing water from the housing,

wherein the water outlet pipe defines a steam discharge hole at an upward facing circumferential surface of the water outlet pipe, the steam discharge hole being configured to transmit steam generated by the pump to the steam supplier,

wherein the water outlet pipe comprises:

a coupling region that defines the steam discharge hole,

a first portion that extends from the coupling region toward the housing, and

a second portion that extends from the coupling region in a direction away from the housing, and

wherein a cross-sectional area of the first portion is less than a cross-sectional area of the second portion.

15. The dishwasher of claim **14**, wherein the water outlet pipe further comprises a stepped portion that protrudes from an inner surface of the water outlet pipe and that is disposed at the first portion of the water outlet pipe.

16. The dishwasher of claim **14**, wherein a thickness of the first portion of the water outlet pipe is greater than a thickness of the second portion of the water outlet pipe.

17. The dishwasher of claim **14**, wherein the first portion of the water outlet pipe is located at an upstream side relative to the coupling region, and

wherein the second portion of the water outlet pipe is located at a downstream side relative to the coupling region.

18. A dishwasher comprising:

a tub;

a spray arm configured to spray washing water into the tub;

a steam supplier configured to discharge steam into the tub;

a sump configured to store washing water; and

a pump that is configured to supply washing water stored in the sump to the spray arm, that is configured to generate steam, and that is configured to supply steam to the steam supplier, the pump comprising:

a housing configured to receive washing water from the sump,

a heater configured to heat washing water in the housing and to generate steam based on heating washing water in the housing,

an impeller disposed in the housing, and

a water outlet pipe that protrudes from the housing, that communicates with the housing, and that is configured to discharge washing water from the housing,

wherein the water outlet pipe defines a steam discharge hole at an upward facing circumferential surface of the water outlet pipe, the steam discharge hole being configured to transmit steam generated by the pump to the steam supplier,

wherein the steam discharge hole is defined at a position above a horizontal axis passing through the circumferential surface of the water outlet pipe, and

wherein a center axis of the steam discharge hole defines an acute angle with respect to the horizontal axis.

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